

Spotly — Features Grouping

Luxury Venue Discovery and Booking Platform

Course: ISS296 — Spring 2026
Instructor: Mr. Amine Ben Hassouna
Team: Cyrine Krichen, Eya Sassi,
Lina Khemiri, Youssef Ben Jemaa

This document maps each application feature to the data model entities it involves. Feature IDs are as defined in the *Project Features & User Roles* document. Entities correspond to the classes in the UML class diagram.

Each row lists the feature identifier, its name, and all data model entities that are created, read, updated, or deleted during the execution of that feature.

Phase 1 — Core Features

ID	Feature Name	Entities Involved
F1	User Authentication	User — A User record is created on registration. On login, the matching User is read and a session object is written to local storage.
F2	Venue Discovery and Search	Venue — The full VENUE_LIST is read and filtered client-side. No write operations occur.
F3	Venue Public Profile	Venue , Environment , Review — The Venue record is read for identity details and the slideshow. Environment records are read to list available spaces. Review records are read to display the rating summary and guest feedback.
F4	Digital Menu	Venue , MenuItem , Environment — The Venue and Environment records are read for context (venue name, environment name). MenuItem records are read and filtered by environmentIds to show the correct items for the scanned table.
F5	Booking: Environment Selection	Venue , Environment — Venue is read to display the venue header. All Environment records for the venue are read to render the selection cards.

Continued on next page. . .

ID	Feature Name	Entities Involved
F6	Booking: Seat and Table Selection	Environment, Reservation — The selected Environment (including its FloorElement list) is read to render the floor plan. Existing Reservation records are read to derive live table availability (available vs. reserved).
F7	Reservation Submission	Reservation, ReservationLog — A new Reservation record is created with status REQUESTED . A corresponding ReservationLog entry is created to record the initial status transition.
F8	Real-Time Reservation Status Tracking	Reservation — The target Reservation record is read and watched via the local storage storage event. The UI reacts to status changes (REQUESTED → APPROVED / REJECTED) without additional writes.
F9	Client Dashboard and Reservation Management	Reservation, Review — All Reservation records for the current user are read and segmented into upcoming and past lists. On cancellation, the matching Reservation status is updated to CANCELLED . Review records are read to show which past visits have already been reviewed.
F10	Guest Review Submission	Review, Reservation — A new Review record is created linked to the completed Reservation . The Reservation is read to verify eligibility (status must be COMPLETED).
F11	Admin Venue Onboarding	Venue, Environment — A new Venue record is created (or an existing one is updated) with the admin's identity details. At least one Environment record is created to complete onboarding.
F12	Admin Dashboard and Overview	Venue, Reservation, ReservationLog — The admin's Venue record is read for the greeting. Reservation records are aggregated to compute statistics. ReservationLog records are read for trend data.

Continued on next page. . .

ID	Feature Name	Entities Involved
F13	Floor Plan Builder	Environment, FloorElement — Environment records are created, updated, and deleted. The embedded FloorElement array within each Environment is modified as elements are added, moved, resized, or removed.
F14	Menu Management	MenuItem, Environment — MenuItem records are created, updated (name, price, description, tags, allergens, availability), and deleted. Environment records are read to populate the environment scope selector.
F15	Reservation Queue Management	Reservation, ReservationLog — All Reservation records for the venue are read and filtered. On approval or rejection, the target Reservation status is updated and a new ReservationLog entry is created to record the admin action.
F16	Guest Review Management	Review — All Review records for the venue are read. On moderation, the status field of the target Review is toggled between VISIBLE and HIDDEN .
F17	QR Code Generation	Venue, Environment, FloorElement — The Venue and all its Environment records are read. The FloorElement list is traversed to generate one QR code per table element. No records are written.
F18	Venue Identity and Settings	Venue — The admin's Venue record is read into the editor form. On save, the Venue record is updated with new identity fields, image URLs, and hour configurations.

Continued on next page...

ID	Feature Name	Entities Involved
F19	Staff Operations Dashboard	Reservation, ReservationLog, VenueStaff, WaiterCall — VenueStaff is read to verify access and determine the associated venue. Reservation records are read and filtered by date and environment for the floor map. Status transitions (check-in, check-out, no-show, walk-in) update the Reservation and create ReservationLog entries. WaiterCall records are read for live alerts and updated on acknowledgement.
F20	Waiter Call System	WaiterCall, Environment — A new WaiterCall record is created when a guest taps the call button on the digital menu. The Environment record is read to populate the call with the environment name and element label. On staff acknowledgement, the WaiterCall status is updated to acknowledged .

Phase 2 — AI-Powered Features

ID	Feature Name	Entities Involved
F21	AI Voice Booking	Reservation, Venue — Venue records are read and sent to the AI as context for venue matching. The AI-parsed result is used to pre-populate a new Reservation in the confirmation form, which is then submitted using the standard F7 flow.
F22	AI Semantic Venue Discovery	Venue — The full VENUE_LIST (id, name, description, ambienceTags, activities, dressCode) is read and forwarded to the AI. The AI returns a ranked list of venue IDs. No records are written.

ID	Feature Name	Entities Involved
F23	AI Reservation Insights	Reservation, ReservationLog — Reservation records for the past 30 days are aggregated (totals, status breakdown, day-of-week distribution, hour distribution, average guest count). ReservationLog entries may be consulted for trend context. The aggregated statistics are forwarded to the AI; no records are written.